800 V-CLASS SIC TRACTION INVERTER CONTROLLER - 150 KW CONTINUOUS / 220 KW PEAK

SYS.3 System Architectural Design & Detailed HW/SW Functional Requirements

Rev B0 - Detailed Block Package

Requirement authority: Traction Controller OEM Design & Development Baseline Rev D1 + TRC-SYS-REF-001 Rev B0 | Freeze/authority: 31-Aug-2025

Document Field	Controlled Value
Document ID	TRC-SYS3-ARCH-REQ-002
Revision	Rev B0 - Detailed Block Package
Technical information freeze / authority	31-Aug-2025
Primary source baseline	Traction Controller OEM Design & Development Baseline Rev D1 + TRC-SYS-REF-001 Rev B0
Process scope	Automotive SPICE 4.0 SYS.3 with downstream HWE.1 / SWE.1 / SYS.4 / SYS.5 / VAL.1 handoff
Architecture blocks	18
Detailed SYS.3 requirements	90
Claim boundary	Derived SYS.3 engineering work product; released OEM requirements and vehicle safety case take precedence.

1. Purpose, Scope and Requirement Authority

This package converts the controlled product baseline into a block-by-block SYS.3 architectural design with explicit hardware/software allocation, algorithms, safety mechanisms, interfaces, verification scenarios and traceability handoff.

The document and companion workbook are intended to be baselined together. A populated trace link or planned test is not executed evidence; closure requires an approved procedure, controlled test configuration, raw result, pass/fail decision and problem/deviation disposition.

ASIL and cybersecurity classifications remain inherited from the approved vehicle/item HARA, FSC/TSC and TARA. This architecture does not create a final ASIL or homologation claim.

2. Frozen / Controlled Technical Baseline

- 800 V-class SiC traction inverter controller.
- Reference power class 150 kW continuous / 220 kW for 10 s peak.
- Four-quadrant operation across 250-920 V battery envelope.
- Primary control FOC with SVPWM/DPWM; SPWM retained for commissioning/reference unless separately released.
- Torque tracking target +/-2% or +/-3 Nm with torque reference update <=5 ms.
- Resolver angle error <=+/-0.5 electrical degree; hard fault detection <=2 ms.
- Phase current measurement <=+/-1.0% FS over -40 to +105 degC; DC-link measurement <=+/-0.5% reading from 200-920 V.
- Qualified DESAT/short-circuit gate-disable initiation <=2 us; independent torque monitoring <=1 ms nominal.
- 15-year/300,000 km aging assumption; automated EOL calibration, gate diagnostics and serialized traceability.

3. High-Level System Architecture

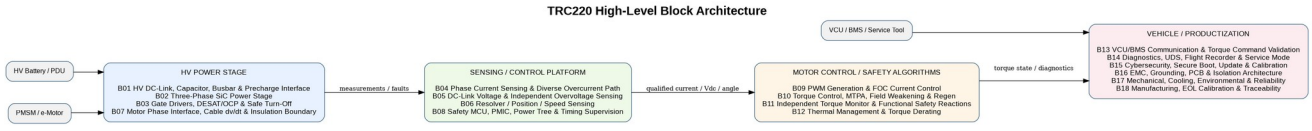


Figure 1 - High-Level Block Architecture

4. End-to-End Algorithm / Information Flow

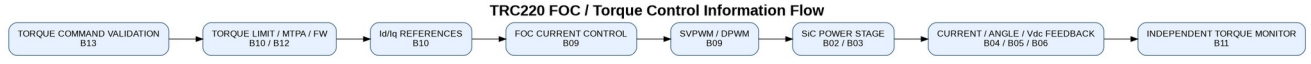


Figure 2 - Control / Algorithm Information Flow

5. Operating States and Safety Flow

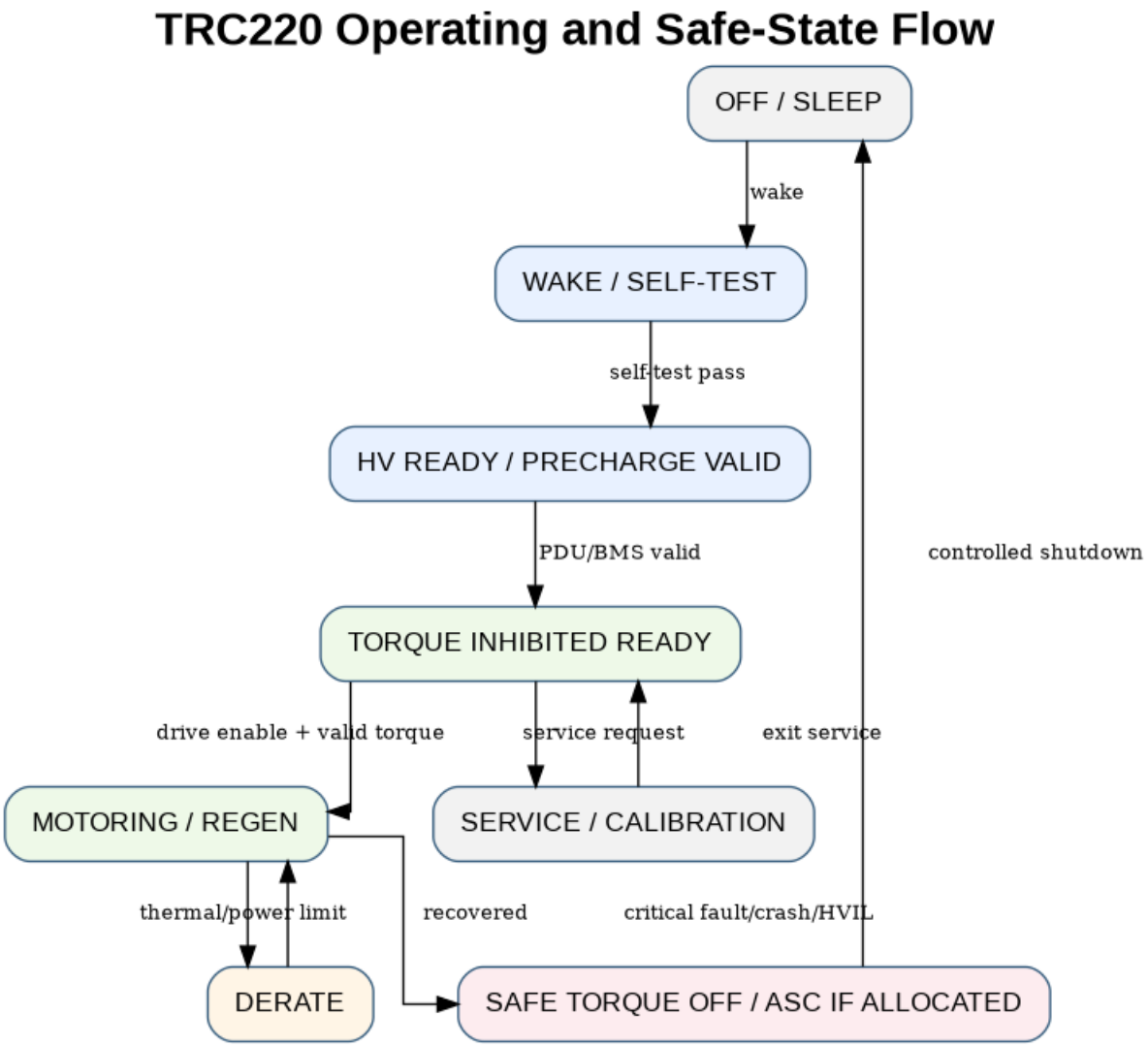


Figure 3 - Operating / Safe-State Flow

6. B01 - HV DC-Link, Capacitor, Busbar & Precharge Interface

800 V-class HV input, DC-link film capacitor, busbar, local HF decoupling and PDU/precharge interface.

Architecture and boundaries

- DC-link voltage envelope: The traction inverter HV interface shall support four-quadrant operation across the released 250-920 V vehicle battery envelope while the precision Vdc sensing covers at least 200-920 V. Acceptance / boundary: No commanded operation outside released voltage/derating envelope; UV/OV reactions defined.
- DC-link capacitor energy and ripple: Film capacitor and busbar shall be sized for ripple current, regen transients, peak-power pulses, ESR/ESL, hotspot and mission-profile life. Acceptance / boundary: 150 kW continuous / 220 kW 10 s peak reference duty closes thermal/lifetime analysis.

- Commutation loop inductance: DC-link busbar/local decoupling shall minimize commutation-loop inductance and keep Vds overshoot within the validated device/module envelope. Acceptance / boundary: DPT and dyno switching waveforms remain within released Vds/ringing/EMC limits.

Algorithm / dynamic behavior

DC-link supervisor: acquire Vdc -> validate range/ripple -> BMS/PDU state -> enable gate permit -> monitor OV/UV/regen energy -> discharge status.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B01-01	DC-link voltage envelope	The traction inverter HV interface shall support four-quadrant operation across the released 250-920 V vehicle battery envelope while the precision Vdc sensing covers at least 200-920 V.	No commanded operation outside released voltage/derating envelope; UV/OV reactions defined.	HW/SW	Source-derived	MR-TRC-002;MR-TRC-006
TRC220-SYS3-B01-02	DC-link capacitor energy and ripple	Film capacitor and busbar shall be sized for ripple current, regen transients, peak-power pulses, ESR/ESL, hotspot and mission-profile life.	150 kW continuous / 220 kW 10 s peak reference duty closes thermal/lifetime analysis.	HW	Engineering-derived	MR-TRC-005;MR-TRC-027
TRC220-SYS3-B01-03	Commutation loop inductance	DC-link busbar/local decoupling shall minimize commutation-loop inductance and keep Vds overshoot within the validated device/module envelope.	DPT and dyno switching waveforms remain within released Vds/ringing/EMC limits.	HW	Engineering-derived	MR-TRC-003;MR-TRC-008
TRC220-SYS3-B01-04	Precharge/ contactor interlock	PWM/gate enable shall remain inhibited until valid PDU/BMS precharge/contactor state and Vdc plausibility are confirmed.	No inverter switching with incomplete/invalid vehicle precharge state.	HW/SW	Engineering-derived	MR-TRC-024
TRC220-SYS3-B01-05	Regen overvoltage containment	Regen torque shall be reduced/inhibited using BMS charge-power/current limits and independent Vdc OV protection before hazardous DC-link rise.	No uncontrolled Vdc rise during BMS limit reduction/network timeout.	HW/SW	Source-derived	MR-TRC-010

Safety, diagnostics and failure containment

- DC-link voltage envelope: No commanded operation outside released voltage/derating envelope; UV/OV reactions defined.
- DC-link capacitor energy and ripple: 150 kW continuous / 220 kW 10 s peak reference duty closes thermal/lifetime analysis.

- Commutation loop inductance: DPT and dyno switching waveforms remain within released Vds/ringing/EMC limits.
- Precharge/contactor interlock: No inverter switching with incomplete/invalid vehicle precharge state.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-01 - Infineon EV traction inverter application and 800 V system solution (Infineon): source link
- REF-TRC-02 - TIDM-02014 800 V 300 kW SiC traction inverter reference design (Texas Instruments / Wolfspeed): source link

7. B02 - Three-Phase SiC Power Stage

Three-phase 1200 V-class SiC bridge sized for 150 kW continuous / 220 kW 10 s peak reference power.

Architecture and boundaries

- Power rating: The three-phase SiC power stage shall support 150 kW continuous and 220 kW for 10 s reference peak duty subject to released coolant, battery voltage and motor operating point. Acceptance / boundary: Thermal/electrical limits demonstrated on dyno for continuous and 10 s peak duty.
- SiC voltage class: The 800 V-class design shall use a 1200 V-class automotive SiC module/device family unless a released voltage/transient study justifies another class. Acceptance / boundary: Static/transient Vds margin including overshoot and aging closes.
- Phase-leg shoot-through prevention: Deadtime, PWM interlock, gate-driver logic and fault paths shall prevent destructive simultaneous conduction in defined single faults. Acceptance / boundary: No destructive shoot-through during PWM/reset/driver fault injection.

Algorithm / dynamic behavior

Power-stage actuation follows synchronized PWM and gate-enable only; losses and junction estimation feed torque capability.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B02-01	Power rating	The three-phase SiC power stage shall support 150 kW continuous and 220 kW for 10 s reference peak duty subject to released coolant, battery voltage and motor operating point.	Thermal/electrical limits demonstrated on dyno for continuous and 10 s peak duty.	HW/SW	Source-derived	MR-TRC-001;MR-TRC-005
TRC220-SYS3-B02-02	SiC voltage class	The 800 V-class design shall use a 1200 V-class automotive SiC module/device	Static/transient Vds margin including overshoot and aging closes.	HW	Engineering-derived	MR-TRC-003

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		family unless a released voltage/transient study justifies another class.				
TRC220-SYS3-B02-03	Phase-leg shoot-through prevention	Deadtime, PWM interlock, gate-driver logic and fault paths shall prevent destructive simultaneous conduction in defined single faults.	No destructive shoot-through during PWM/reset/driver fault injection.	HW/SW	Engineering-derived	MR-TRC-003
TRC220-SYS3-B02-04	Reverse conduction / regen capability	Power stage device and cooling selection shall support four-quadrant current including reverse conduction/regen loss cases.	Motoring and regen loss/thermal maps released across voltage/current corners.	HW	Engineering-derived	MR-TRC-002
TRC220-SYS3-B02-05	Power-module health margins	Module temperature, solder/sinter/interconnect cycling and current density shall remain within released lifetime constraints.	Mission-profile damage accumulation supports 15-year/300,000 km target.	HW	Engineering-derived	MR-TRC-027

Safety, diagnostics and failure containment

- SiC voltage class: Static/transient Vds margin including overshoot and aging closes.
- Phase-leg shoot-through prevention: No destructive shoot-through during PWM/reset/driver fault injection.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-01 - Infineon EV traction inverter application and 800 V system solution (Infineon): source link
- REF-TRC-02 - TIDM-02014 800 V 300 kW SiC traction inverter reference design (Texas Instruments / Wolfspeed): source link

8. B03 - Gate Drivers, DESAT/OCP & Safe Turn-Off

Isolated drivers, DESAT/OCP, Miller control, soft-off, UVLO, fault reporting and independent gate inhibit.

Architecture and boundaries

- DESAT/SC reaction: After a qualified DESAT/short-circuit event, gate disable shall initiate within <=2 us and the turn-off profile shall keep device stress within validated limits. Acceptance / boundary: Gate-disable initiation <=2 us; DPT/SC evidence shows no destructive stress.

- Safe default off: All gate outputs shall default disabled during reset, brownout, driver UVLO and unpowered-driver conditions. Acceptance / boundary: No uncontrolled gate-on state in normal power sequencing or defined single faults.
- Tunable gate profile: Gate resistance, clamp, soft turn-off and optional negative-bias settings shall be tunable by released hardware/calibration for module-specific DPT optimization. Acceptance / boundary: All profiles remain inside device Vgs/Vds/EMC/motor dv/dt limits.

Algorithm / dynamic behavior

Fast fault path: DESAT/OC -> hardware inhibit <= allocated FTTI -> safe turn-off -> latch -> safety manager/diagnostics.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B03-01	DESAT/SC reaction	After a qualified DESAT/short-circuit event, gate disable shall initiate within <=2 us and the turn-off profile shall keep device stress within validated limits.	Gate-disable initiation <=2 us; DPT/SC evidence shows no destructive stress.	HW	Source-derived	MR-TRC-003
TRC220-SYS3-B03-02	Safe default off	All gate outputs shall default disabled during reset, brownout, driver UVLO and unpowered-driver conditions.	No uncontrolled gate-on state in normal power sequencing or defined single faults.	HW	Source-derived	MR-TRC-009
TRC220-SYS3-B03-03	Tunable gate profile	Gate resistance, clamp, soft turn-off and optional negative-bias settings shall be tunable by released hardware/calibration for module-specific DPT optimization.	All profiles remain inside device Vgs/Vds/EMC/motor dv/dt limits.	HW/SW	Source-derived	MR-TRC-020
TRC220-SYS3-B03-04	Miller / false-turn-on control	Driver and layout shall provide adequate Miller/false-turn-on margin at maximum dv/dt and common-mode transient.	No false turn-on in DPT/CM immunity tests.	HW	Engineering-derived	MR-TRC-008
TRC220-SYS3-B03-05	Independent gate inhibit	Crash/HVIL/ critical safety input shall remove gate permission through an independent path where allocated by TSC.	Application CPU/software cannot defeat allocated independent inhibit.	HW/SW	Source-derived	MR-TRC-024

Safety, diagnostics and failure containment

- DESAT/SC reaction: Gate-disable initiation <=2 us; DPT/SC evidence shows no destructive stress.
- Safe default off: No uncontrolled gate-on state in normal power sequencing or defined single faults.
- Tunable gate profile: All profiles remain inside device Vgs/Vds/EMC/motor dv/dt limits.
- Miller / false-turn-on control: No false turn-on in DPT/CM immunity tests.

Verification / DVP focus

- nominal operating point

- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-03 - AURIX TC3xx main inverter functional safety reference (Infineon): source link
- REF-TRC-04 - Automotive isolated gate drivers for traction inverters (Infineon): source link

9. B04 - Phase Current Sensing & Diverse Overcurrent Path

High-bandwidth phase-current acquisition, calibration, saturation diagnostics and independent/dissimilar OC plausibility.

Architecture and boundaries

- Phase-current accuracy: Phase-current measurement shall meet <=+/-1.0% FS worst-case over -40 to +105 degC for the released current range. Acceptance / boundary: WCCA + chamber/calibration test meets <=+/-1.0% FS.
- PWM synchronized sampling: Current sampling shall be synchronized to PWM and switching-noise windows to provide stable FOC feedback. Acceptance / boundary: No control instability or excessive sample corruption across modulation region.
- Offset/gain qualification: Current offset/gain adaptation shall only occur in qualified states and remain bounded by immutable limits. Acceptance / boundary: Adaptive correction cannot hide a sensor fault or exceed WCCA.

Algorithm / dynamic behavior

Current acquisition: synchronized sample -> offset/gain correction -> Clarke transform -> plausibility -> Id/Iq feedback; diverse OC bypasses normal FOC.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B04-01	Phase-current accuracy	Phase-current measurement shall meet <=+/-1.0% FS worst-case over -40 to +105 degC for the released current range.	WCCA + chamber/calibration test meets <=+/-1.0% FS.	HW/SW	Source-derived	MR-TRC-007
TRC220-SYS3-B04-02	PWM synchronized sampling	Current sampling shall be synchronized to PWM and switching-noise windows to provide stable FOC feedback.	No control instability or excessive sample corruption across modulation region.	HW/SW	Engineering-derived	MR-TRC-007
TRC220-SYS3-B04-03	Offset/gain qualification	Current offset/gain adaptation shall only occur in qualified states and remain bounded by immutable limits.	Adaptive correction cannot hide a sensor fault or exceed WCCA.	SW	Engineering-derived	MR-TRC-007
TRC220-SYS3-B04-04	Diverse overcurrent	A diverse hardware/driver OC path shall protect against sensor/CPU/control	Qualified OC fault causes hardware-safe reaction independent of FOC task.	HW	Source-derived	MR-TRC-007;MR-TRC-021

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		failure and shall not depend only on the precision ADC path.				
TRC220-SYS3-B04-05	Sensor plausibility	Independent/ dissimilar phase-current plausibility shall detect offset, saturation, open/short and ground-shift conditions.	No plausible unsafe torque from single current-sensor fault.	HW/SW	Source-derived	MR-TRC-021

Safety, diagnostics and failure containment

- Phase-current accuracy: WCCA + chamber/calibration test meets <=+/-1.0% FS.
- Offset/gain qualification: Adaptive correction cannot hide a sensor fault or exceed WCCA.
- Diverse overcurrent: Qualified OC fault causes hardware-safe reaction independent of FOC task.
- Sensor plausibility: No plausible unsafe torque from single current-sensor fault.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-02 - TIDM-02014 800 V 300 kW SiC traction inverter reference design (Texas Instruments / Wolfspeed): source link

10. B05 - DC-Link Voltage & Independent Overvoltage Sensing

Precision Vdc acquisition from 200-920 V plus independently qualified hazardous overvoltage path.

Architecture and boundaries

- Vdc accuracy: The precision DC-link measurement shall meet <=+/-0.5% reading from 200-920 V over the released temperature/aging envelope. Acceptance / boundary: Chamber/WCCA/calibration meets <=+/-0.5% reading.
- Independent OV protection: Hazardous overvoltage shall be independently detected by a path sufficiently diverse from the main precision software signal. Acceptance / boundary: OV reaction remains effective with main CPU task blocked or precision channel invalid.
- Divider stress/insulation: HV divider/isolator shall meet voltage coefficient, pulse energy, creepage/clearance and isolation-lifetime limits. Acceptance / boundary: WCCA/insulation review closed at 920 V + transient envelope.

Algorithm / dynamic behavior

Vdc manager: filtered precision value -> plausibility -> BMS/regen manager + independent OV comparator/status -> torque/inhibit reaction.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B05-01	Vdc accuracy	The precision DC-link measurement shall meet <=+/-0.5% reading from 200-920 V over the released temperature/aging envelope.	Chamber/WCCA/ calibration meets <=+/-0.5% reading.	HW/SW	Source-derived	MR-TRC-006
TRC220-SYS3-B05-02	Independent OV protection	Hazardous overvoltage shall be independently detected by a path sufficiently diverse from the main precision software signal.	OV reaction remains effective with main CPU task blocked or precision channel invalid.	HW	Source-derived	MR-TRC-006;MR-TRC-021
TRC220-SYS3-B05-03	Divider stress/insulation	HV divider/isolator shall meet voltage coefficient, pulse energy, creepage/clearance and isolation-lifetime limits.	WCCA/insulation review closed at 920 V + transient envelope.	HW	Engineering-derived	MR-TRC-006
TRC220-SYS3-B05-04	Regen limit integration	Vdc rate and absolute level shall feed regen/torque reduction so BMS charge limits and DC-link protection act coherently.	No oscillatory torque/Vdc hunting during BMS limit changes.	SW	Engineering-derived	MR-TRC-010
TRC220-SYS3-B05-05	Sensor disagreement reaction	Diverse Vdc channels shall have calibrated disagreement bands and conservative reaction.	Disagreement never increases allowed regen/motoring power.	SW	Engineering-derived	MR-TRC-021

Safety, diagnostics and failure containment

- Vdc accuracy: Chamber/WCCA/calibration meets <=+/-0.5% reading.
- Independent OV protection: OV reaction remains effective with main CPU task blocked or precision channel invalid.
- Divider stress/insulation: WCCA/insulation review closed at 920 V + transient envelope.
- Regen limit integration: No oscillatory torque/Vdc hunting during BMS limit changes.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-02 - TIDM-02014 800 V 300 kW SiC traction inverter reference design (Texas Instruments / Wolfspeed): source link

11. B06 - Resolver / Position / Speed Sensing

Primary resolver interface, excitation/AFE, angle computation, plausibility and diverse monitor strategy.

Architecture and boundaries

- Resolver angle accuracy: Resolver-based electrical angle error shall remain <=+/-0.5 electrical degree over released speed/temperature/supply conditions. Acceptance / boundary: Dyno/fixture angle comparison meets <=+/-0.5 electrical degree.
- Hard-fault detection: Resolver excitation/sin/cos/ADC/processing hard faults shall be detected within <=2 ms or the released equivalent safety mechanism timing. Acceptance / boundary: Open/short/frozen/invalid excitation faults detected <=2 ms.
- Offset/alignment calibration: Electrical zero/offset calibration shall be controlled, bounded and serialized at EOL. Acceptance / boundary: Wrong/missing calibration inhibits torque or applies safe fallback.

Algorithm / dynamic behavior

Resolver path: excitation -> sin/cos acquisition -> demod/angle -> electrical offset -> speed -> plausibility; observer/HFI initially monitor unless separately safety-qualified.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B06-01	Resolver angle accuracy	Resolver-based electrical angle error shall remain <=+/-0.5 electrical degree over released speed/temperature/supply conditions.	Dyno/fixture angle comparison meets <=+/-0.5 electrical degree.	HW/SW	Source-derived	MR-TRC-004
TRC220-SYS3-B06-02	Hard-fault detection	Resolver excitation/sin/cos/ADC/processing hard faults shall be detected within <=2 ms or the released equivalent safety mechanism timing.	Open/short/frozen/invalid excitation faults detected <=2 ms.	HW/SW	Source-derived	MR-TRC-004
TRC220-SYS3-B06-03	Offset/alignment calibration	Electrical zero/offset calibration shall be controlled, bounded and serialized at EOL.	Wrong/missing calibration inhibits torque or applies safe fallback.	HW/SW	Engineering-derived	MR-TRC-030
TRC220-SYS3-B06-04	Diverse position monitor	Resolver remains primary authority; HFI/observer shall initially be a diverse monitor unless separately safety-qualified.	Observer disagreement detected; observer alone does not silently replace failed resolver.	SW	Source-derived	MR-TRC-021
TRC220-SYS3-B06-05	Speed plausibility	Speed shall be derived/qualified with rate and direction checks and cross-compared to vehicle/motor expectations where available.	Implausible speed cannot produce unsafe field-weakening or torque.	SW	Engineering-derived	MR-TRC-021

Safety, diagnostics and failure containment

- Resolver angle accuracy: Dyno/fixture angle comparison meets <=+/-0.5 electrical degree.
- Hard-fault detection: Open/short/frozen/invalid excitation faults detected <=2 ms.
- Offset/alignment calibration: Wrong/missing calibration inhibits torque or applies safe fallback.
- Diverse position monitor: Observer disagreement detected; observer alone does not silently replace failed resolver.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-05 - Design priorities in EV traction inverters - resolver/front-end considerations (Texas Instruments): source link

12. B07 - Motor Phase Interface, Cable dv/dt & Insulation Boundary

Motor terminals, phase busbars/cables, common-mode behavior and motor insulation dv/dt boundary.

Architecture and boundaries

- Motor dv/dt boundary: Motor supplier repetitive terminal dv/dt / insulation withstand boundary shall be available before final gate-slew release. Acceptance / boundary: DPT + motor/cable test stays inside released repetitive dv/dt/overshoot envelope.
- Phase-cable CM behavior: Motor cable length/shield/grounding and phase busbar shall be included in EMC/common-mode validation. Acceptance / boundary: No unsafe control/sensor behavior in worst-case representative cable installation.
- Phase terminal insulation: Phase terminals, busbars and connectors shall meet working/transient voltage, creepage/clearance, vibration and temperature limits. Acceptance / boundary: Insulation/mechanical DVP passes at aged/hot conditions.

Algorithm / dynamic behavior

No central real-time algorithm; gate-slew profile is constrained by motor terminal dv/dt, EMC, Vds/Vgs and thermal evidence.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B07-01	Motor dv/dt boundary	Motor supplier repetitive terminal dv/dt / insulation withstand boundary shall be available before final gate-slew release.	DPT + motor/cable test stays inside released repetitive dv/dt/overshoot envelope.	HW/SW	Source-derived	MR-TRC-020
TRC220-SYS3-B07-02	Phase-cable CM behavior	Motor cable length/shield/grounding and phase busbar shall be	No unsafe control/sensor behavior in worst-case representative	HW	Engineering-derived	MR-TRC-008

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		included in EMC/common-mode validation.	cable installation.			
TRC220-SYS3-B07-03	Phase terminal insulation	Phase terminals, busbars and connectors shall meet working/transient voltage, creepage/clearance, vibration and temperature limits.	Insulation/ mechanical DVP passes at aged/hot conditions.	HW	Engineering-derived	MR-TRC-026
TRC220-SYS3-B07-04	Gate profile constraint	Adaptive/fixed gate profiles shall remain inside motor insulation, EMC, Vds, Vgs and thermal limits at every released operating point.	Production map contains no unvalidated slew-rate cell.	HW/SW	Source-derived	MR-TRC-020
TRC220-SYS3-B07-05	Open-phase / connection fault	Open/abnormal phase behavior shall be detected through current/voltage/torque plausibility and result in safe torque reaction.	No sustained unsafe torque under defined phase open/connection faults.	HW/SW	Engineering-derived	MR-TRC-021

Safety, diagnostics and failure containment

- Motor dv/dt boundary: DPT + motor/cable test stays inside released repetitive dv/dt/overshoot envelope.
- Phase-cable CM behavior: No unsafe control/sensor behavior in worst-case representative cable installation.
- Phase terminal insulation: Insulation/mechanical DVP passes at aged/hot conditions.
- Gate profile constraint: Production map contains no unvalidated slew-rate cell.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-01 - Infineon EV traction inverter application and 800 V system solution (Infineon): source link

13. B08 - Safety MCU, PMIC, Power Tree & Timing Supervision

Real-time MCU, lockstep/ECC, PMIC/watchdog, PWM/ADC timing, MPU and startup/reset supervision.

Architecture and boundaries

- Safety MCU platform: The control platform shall provide the real-time PWM/ADC/resolver resources, lockstep/ECC and safety features needed for 150/220 kW control and monitoring. Acceptance / boundary: Resource and safety architecture review closes with WCET/headroom.

- Startup readiness: The controller shall complete startup self-test and publish ready <=500 ms after valid wake, excluding vehicle precharge time. Acceptance / boundary: Ready <=500 ms with no torque/gate enable before critical test completion.
- Sleep/off safe outputs: Sleep/off shall maintain gate disabled and defined safe outputs with bounded quiescent behavior. Acceptance / boundary: No spurious PWM/gate-enable during sleep entry/exit/brownout.

Algorithm / dynamic behavior

Startup: LV power -> PMIC/reset -> MCU BIST/ECC -> watchdog/MPU -> resolver/current/Vdc/gate diagnostics -> safe PWM init -> torque-enable readiness.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B08-01	Safety MCU platform	The control platform shall provide the real-time PWM/ADC/resolver resources, lockstep/ECC and safety features needed for 150/220 kW control and monitoring.	Resource and safety architecture review closes with WCET/headroom.	HW/SW	Engineering-derived	MR-TRC-022
TRC220-SYS3-B08-02	Startup readiness	The controller shall complete startup self-test and publish ready <=500 ms after valid wake, excluding vehicle precharge time.	Ready <=500 ms with no torque/gate enable before critical test completion.	SW	Source-derived	MR-TRC-017
TRC220-SYS3-B08-03	Sleep/off safe outputs	Sleep/off shall maintain gate disabled and defined safe outputs with bounded quiescent behavior.	No spurious PWM/gate-enable during sleep entry/exit/brownout.	HW/SW	Source-derived	MR-TRC-018
TRC220-SYS3-B08-04	Watchdog/PMIC	Independent watchdog/reset/power supervision shall detect stalled application, bad timing and rail faults and remove torque/gate permission as allocated.	Fault injection meets FTTI and safe-state requirements.	HW/SW	Engineering-derived	MR-TRC-009
TRC220-SYS3-B08-05	MPU/WCET freedom from interference	Safety/control partitions shall have measured WCET and MPU/memory/stack protection sufficient to prevent lower-criticality tasks from corrupting fast torque/safety execution.	CPU overload/stack/MPU fault injection preserves safety monitor and/or triggers safe state.	SW	Engineering-derived	MR-TRC-022

Safety, diagnostics and failure containment

- Safety MCU platform: Resource and safety architecture review closes with WCET/headroom.
- Startup readiness: Ready <=500 ms with no torque/gate enable before critical test completion.

- Sleep/off safe outputs: No spurious PWM/gate-enable during sleep entry/exit/brownout.
- Watchdog/PMIC: Fault injection meets FTTI and safe-state requirements.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-03 - AURIX TC3xx main inverter functional safety reference (Infineon): source link
- REF-TRC-06 - EV Traction Inverter Control Reference Design Gen 3 (NXP): source link

14. B09 - PWM Generation & FOC Current Control

High-frequency current loops, Clarke/Park transforms, current regulators, SVPWM/DPWM and SPWM reference mode.

Architecture and boundaries

- Torque update / FOC rate: Torque reference shall be updated <=5 ms and inner current control shall execute at a frequency consistent with the released PWM/control design. Acceptance / boundary: Torque command update <=5 ms; inner-loop deadline misses not observed in worst-case timing test.
- SVPWM/DPWM primary modulation: Production control shall use validated SVPWM/DPWM; SPWM remains commissioning/reference unless separately released. Acceptance / boundary: Only released modulation modes are production-calibration accessible.
- Current regulator stability: Id/Iq current regulators shall remain stable across base speed, field weakening, DC-link variation and current saturation. Acceptance / boundary: No sustained oscillation or integrator windup at voltage/current limits.

Algorithm / dynamic behavior

FOC: torque/flux refs -> Id/Iq refs -> current feedback -> PI/decoupling -> Vd/Vq saturation -> inverse Park -> SVPWM/DPWM -> synchronized PWM.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B09-01	Torque update / FOC rate	Torque reference shall be updated <=5 ms and inner current control shall execute at a frequency consistent with the released PWM/control design.	Torque command update <=5 ms; inner-loop deadline misses not observed in worst-case timing test.	SW	Source-derived	MR-TRC-001
TRC220-SYS3-B09-02	SVPWM/DPWM primary modulation	Production control shall use validated SVPWM/DPWM; SPWM remains commissioning/ref	Only released modulation modes are production-calibration accessible.	SW	Source-derived	MR-TRC-008

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		erence unless separately released.				
TRC220-SYS3-B09-03	Current regulator stability	Id/Iq current regulators shall remain stable across base speed, field weakening, DC-link variation and current saturation.	No sustained oscillation or integrator windup at voltage/current limits.	SW	Engineering-derived	MR-TRC-011
TRC220-SYS3-B09-04	Sampling/PWM synchronization	PWM generation and current/position acquisition shall use deterministic synchronization and deadtime compensation where required.	Phase-current reconstruction/control error remains inside torque target.	HW/SW	Engineering-derived	MR-TRC-001
TRC220-SYS3-B09-05	Control saturation handling	Voltage-vector saturation and anti-windup shall preserve controllability during low Vdc/high speed and fault-recovery boundaries.	No command discontinuity causing >5% rated torque sign-change step or instability.	SW	Engineering-derived	MR-TRC-002

Safety, diagnostics and failure containment

- SVPWM/DPWM primary modulation: Only released modulation modes are production-calibration accessible.
- Control saturation handling: No command discontinuity causing >5% rated torque sign-change step or instability.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-01 - Infineon EV traction inverter application and 800 V system solution (Infineon): source link
- REF-TRC-02 - TIDM-02014 800 V 300 kW SiC traction inverter reference design (Texas Instruments / Wolfspeed): source link

15. B10 - Torque Control, MTPA, Field Weakening & Regen

Torque arbitration, torque model, MTPA, field weakening, four-quadrant operation and BMS regen limit enforcement.

Architecture and boundaries

- Torque tracking: Requested torque shall be tracked within +/-2% or +/-3 Nm over the released operating region. Acceptance / boundary: Torque error <=max(2%,3 Nm) in defined steady/dynamic test regions.
- Four-quadrant transition: Motoring/regen shall operate across 250-920 V with sign-change discontinuity <=5% rated torque. Acceptance / boundary: Sign reversal discontinuity <=5% rated torque.

- MTPA: Calibrated MTPA/torque-per-ampere optimization shall be used in the base-speed region where valid motor maps exist. Acceptance / boundary: No MTPA map extrapolation outside validated motor data; current remains bounded.

Algorithm / dynamic behavior

Torque manager: requested torque -> validity/limits -> MTPA or FW -> Id/Iq refs -> FOC -> independent torque monitor; regen bounded by BMS charge limits.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B10-01	Torque tracking	Requested torque shall be tracked within +/-2% or +/-3 Nm over the released operating region.	Torque error <=max(2%,3 Nm) in defined steady/dynamic test regions.	SW	Source-derived	MR-TRC-001
TRC220-SYS3-B10-02	Four-quadrant transition	Motoring/regen shall operate across 250-920 V with sign-change discontinuity <=5% rated torque.	Sign reversal discontinuity <=5% rated torque.	SW	Source-derived	MR-TRC-002
TRC220-SYS3-B10-03	MTPA	Calibrated MTPA/torque-per-ampere optimization shall be used in the base-speed region where valid motor maps exist.	No MTPA map extrapolation outside validated motor data; current remains bounded.	SW	Source-derived	MR-TRC-012
TRC220-SYS3-B10-04	Field weakening	Voltage-limited field weakening shall support rated maximum speed without current-regulator instability.	Stable operation across released speed/Vdc envelope with current/voltage limits respected.	SW	Source-derived	MR-TRC-011
TRC220-SYS3-B10-05	BMS regen limits	Regen torque shall obey valid BMS charge-power/current limits; stale/missing limits force conservative local behavior and prevent uncontrolled Vdc rise.	No regen beyond valid BMS/local limits; timeout reaction validated.	SW	Source-derived	MR-TRC-010

Safety, diagnostics and failure containment

- Torque tracking: Torque error <=max(2%,3 Nm) in defined steady/dynamic test regions.
- Four-quadrant transition: Sign reversal discontinuity <=5% rated torque.
- Field weakening: Stable operation across released speed/Vdc envelope with current/voltage limits respected.
- BMS regen limits: No regen beyond valid BMS/local limits; timeout reaction validated.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition

- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-01 - Infineon EV traction inverter application and 800 V system solution (Infineon): source link
- REF-TRC-07 - AM263x for traction inverters / real-time motor control (Texas Instruments): source link

16. B11 - Independent Torque Monitor & Functional Safety Reactions

Independent torque estimation/monitor, gate-enable safety chain, controlled active-short-circuit option and safe-state selection.

Architecture and boundaries

- Independent torque monitor: An independent torque monitor shall execute <=1 ms nominal task period or equivalent safety mechanism. Acceptance / boundary: Monitor timing <=1 ms nominal and independent enough to detect main control fault.
- Torque deviation reaction: Commanded/expected torque shall be cross-checked against independent current/angle/torque estimates with calibrated tolerance and persistence. Acceptance / boundary: Unsafe torque deviation triggers torque inhibit/safe state within allocated FTTI.
- Active short circuit option: Controlled ASC or alternative safe-state strategy shall only be enabled if allocated by vehicle/system safety analysis. Acceptance / boundary: No generic ASC enable; state/phase/current limits validated for released motor/inverter.

Algorithm / dynamic behavior

Safety torque monitor <=1 ms: command -> independent expected torque -> measured/current/angle-based torque estimate -> deviation -> inhibit/ASC/safe state.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B11-01	Independent torque monitor	An independent torque monitor shall execute <=1 ms nominal task period or equivalent safety mechanism.	Monitor timing <=1 ms nominal and independent enough to detect main control fault.	SW	Source-derived	MR-TRC-022
TRC220-SYS3-B11-02	Torque deviation reaction	Commanded/ expected torque shall be cross-checked against independent current/angle/torque estimates with calibrated tolerance and persistence.	Unsafe torque deviation triggers torque inhibit/safe state within allocated FTTI.	SW	Engineering-derived	MR-TRC-022
TRC220-SYS3-B11-03	Active short circuit option	Controlled ASC or alternative safe-state strategy shall only be enabled if allocated by vehicle/system safety analysis.	No generic ASC enable; state/phase/current limits validated for released motor/inverter.	HW/SW	Source-derived	MR-TRC-023
TRC220-SYS3-B11-04	Crash/HVIL reaction	Vehicle HV inhibit/crash/contactor state shall remove gate enable through an independent path	Asynchronous inhibit removes gate permission within TSC timing even with application fault.	HW/SW	Source-derived	MR-TRC-024

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		where allocated.				
TRC220-SYS3-B11-05	Sensor redundancy integration	Independent/ dissimilar current, Vdc and position plausibility shall feed the torque safety monitor and shall never be averaged to hide disagreement.	Single safety-relevant sensor fault cannot create plausible unsafe torque.	HW/SW	Source-derived	MR-TRC-021

Safety, diagnostics and failure containment

- Independent torque monitor: Monitor timing <=1 ms nominal and independent enough to detect main control fault.
- Torque deviation reaction: Unsafe torque deviation triggers torque inhibit/safe state within allocated FTTI.
- Active short circuit option: No generic ASC enable; state/phase/current limits validated for released motor/inverter.
- Crash/HVIL reaction: Asynchronous inhibit removes gate permission within TSC timing even with application fault.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-03 - AURIX TC3xx main inverter functional safety reference (Infineon): source link
- REF-GEN-01 - ISO 26262 road vehicles - functional safety (ISO): source link

17. B12 - Thermal Management & Torque Derating

SiC/module/baseplate/coolant/motor thermal inputs, junction estimation, current/torque capability and hysteretic recovery.

Architecture and boundaries

- Continuous thermal derating: Torque/current capability shall derate continuously as junction, baseplate, coolant or motor/inverter thermal limits are approached. Acceptance / boundary: Hotter or invalid thermal state never increases torque capability.
- Coolant availability: Coolant availability and local thermal measurements shall be inputs to torque capability. Acceptance / boundary: Loss of required cooling produces bounded derate/shutdown before hard limits.
- Non-emergency derate rate: Non-emergency torque derating shall be rate-limited to avoid driver-perceived steps >10% rated torque per 10 ms. Acceptance / boundary: Step/ramp <=10% rated torque per 10 ms except immediate safety shutdown.

Algorithm / dynamic behavior

Thermal capability: sensor validity -> Tj/baseplate/coolant -> loss/thermal model -> current/torque limit -> monotonic derate -> slow recovery.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B12-01	Continuous thermal derating	Torque/current capability shall derate continuously as junction, baseplate, coolant or motor/inverter thermal limits are approached.	Hotter or invalid thermal state never increases torque capability.	HW/SW	Source-derived	MR-TRC-005
TRC220-SYS3-B12-02	Coolant availability	Coolant availability and local thermal measurements shall be inputs to torque capability.	Loss of required cooling produces bounded derate/shutdown before hard limits.	HW/SW	Source-derived	MR-TRC-019
TRC220-SYS3-B12-03	Non-emergency derate rate	Non-emergency torque derating shall be rate-limited to avoid driver-perceived steps >10% rated torque per 10 ms.	Step/ramp <=10% rated torque per 10 ms except immediate safety shutdown.	SW	Source-derived	MR-TRC-013
TRC220-SYS3-B12-04	Thermal recovery hysteresis	Increasing torque after cooling shall use hysteresis and a slower recovery rate to avoid oscillation.	No cyclic torque hunting around thermal boundaries.	SW	Engineering-derived	MR-TRC-005
TRC220-SYS3-B12-05	Prognostic boundary	Thermal prognostics may request earlier derating/service but shall never raise deterministic safety limits.	Model fault cannot increase hard torque/current capability.	SW	Source-derived	MR-TRC-005

Safety, diagnostics and failure containment

- Continuous thermal derating: Hotter or invalid thermal state never increases torque capability.
- Coolant availability: Loss of required cooling produces bounded derate/shutdown before hard limits.
- Prognostic boundary: Model fault cannot increase hard torque/current capability.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-01 - Infineon EV traction inverter application and 800 V system solution (Infineon): source link

18. B13 - VCU/BMS Communication & Torque Command Validation

CAN FD/E2E command and limit interfaces, freshness, plausibility, BMS regen constraints and vehicle inhibit states.

Architecture and boundaries

- Torque command validity: Stale, invalid or implausible torque commands shall be rejected; E2E protections shall be applied where vehicle architecture allocates them. Acceptance / boundary: Invalid command cannot create torque; timeout/rate/range checks verified.
- BMS limit validity: BMS discharge/charge limits shall be freshness/range/plausibility checked before use. Acceptance / boundary: Stale/missing limit causes conservative local limit, not optimistic default.
- VCU inhibit states: Drive enable, gear/state, crash/HV and service states shall be plausibility-checked before torque is enabled. Acceptance / boundary: Torque enable requires consistent vehicle state and independent safety permits.

Algorithm / dynamic behavior

Command manager: receive -> CRC/counter/freshness -> range/rate/state plausibility -> arbitration -> torque reference; stale/invalid -> conservative local limit.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B13-01	Torque command validity	Stale, invalid or implausible torque commands shall be rejected; E2E protections shall be applied where vehicle architecture allocates them.	Invalid command cannot create torque; timeout/rate/range checks verified.	SW	Source-derived	MR-TRC-014
TRC220-SYS3-B13-02	BMS limit validity	BMS discharge/charge limits shall be freshness/range/plausibility checked before use.	Stale/missing limit causes conservative local limit, not optimistic default.	SW	Engineering-derived	MR-TRC-010
TRC220-SYS3-B13-03	VCU inhibit states	Drive enable, gear/state, crash/HV and service states shall be plausibility-checked before torque is enabled.	Torque enable requires consistent vehicle state and independent safety permits.	SW	Engineering-derived	MR-TRC-024
TRC220-SYS3-B13-04	CAN FD timing	Critical command/limit publication and reception shall meet released cycle/latency/jitter budget under maximum bus load.	No deadline/freshness violation at validated network load.	HW/SW	Engineering-derived	MR-TRC-014
TRC220-SYS3-B13-05	Network fault containment	Diagnostic/ noncritical traffic shall not starve torque/safety communication or control CPU time.	Stress/DoS-style test preserves torque safety deadlines or triggers safe state.	HW/SW	Engineering-derived	MR-TRC-014

Safety, diagnostics and failure containment

- Torque command validity: Invalid command cannot create torque; timeout/rate/range checks verified.
- BMS limit validity: Stale/missing limit causes conservative local limit, not optimistic default.
- VCU inhibit states: Torque enable requires consistent vehicle state and independent safety permits.
- CAN FD timing: No deadline/freshness violation at validated network load.

Verification / DVP focus

- nominal operating point

- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-GEN-03 - ISO 14229 unified diagnostic services (ISO): source link

19. B14 - Diagnostics, UDS, Flight Recorder & Service Mode

DTC/freeze-frame, high-rate flight recorder, UDS routines and torque-inhibited service/calibration mode.

Architecture and boundaries

- UDS diagnostics: The inverter shall support ISO 14229 UDS diagnostics, freeze frame, service routines and DTC life-cycle policy. Acceptance / boundary: Required service functions pass diagnostic conformance/integration tests.
- Flight recorder: A rolling high-rate recorder shall capture torque request, currents, Vdc, angle, gate/driver and safety status around critical faults. Acceptance / boundary: Pre/post trigger data retained for released event classes.
- Service/calibration state: Service/calibration mode shall default to torque generation inhibited and require controlled prerequisites for any energized test. Acceptance / boundary: No unintended torque from diagnostic/calibration session.

Algorithm / dynamic behavior

Diagnostic flow: qualify -> DTC -> freeze-frame/recorder -> persist -> service policy; torque generation remains inhibited by default in service mode.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B14-01	UDS diagnostics	The inverter shall support ISO 14229 UDS diagnostics, freeze frame, service routines and DTC life-cycle policy.	Required service functions pass diagnostic conformance/integration tests.	SW	Source-derived	MR-TRC-015
TRC220-SYS3-B14-02	Flight recorder	A rolling high-rate recorder shall capture torque request, currents, Vdc, angle, gate/driver and safety status around critical faults.	Pre/post trigger data retained for released event classes.	SW	Source-derived	MR-TRC-015
TRC220-SYS3-B14-03	Service/calibration state	Service/calibration mode shall default to torque generation inhibited and require controlled prerequisites for any energized test.	No unintended torque from diagnostic/calibration session.	SW	Source-derived	MR-TRC-025
TRC220-SYS3-B14-04	Fault counters	Operating hours, peak events, short-circuit/thermal/OV events and key health counters shall be retained within NVM	Data survives power cycle and rollover is controlled.	SW	Engineering-derived	MR-TRC-015

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		endurance budget.				
TRC220-SYS3-B14-05	Logging independence	Diagnostic/NVM failure shall not delay fast torque/gate safety reaction.	Safety path works with logging disabled/full/corrupt.	SW	Engineering-derived	MR-TRC-015

Safety, diagnostics and failure containment

- Service/calibration state: No unintended torque from diagnostic/calibration session.
- Logging independence: Safety path works with logging disabled/full/corrupt.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-GEN-03 - ISO 14229 unified diagnostic services (ISO): source link

20. B15 - Cybersecurity, Secure Boot, Update & Calibration

HSM/root-of-trust, authenticated software/calibration, rollback protection and variant coding.

Architecture and boundaries

- Secure boot: Unsigned or modified production software shall be blocked and torque/gate enable shall remain disabled. Acceptance / boundary: Negative security test blocks unsigned image.
- Safety calibration protection: Unauthorized changes to safety/torque/gate calibration shall be rejected using signature/authentication plus immutable hard bounds. Acceptance / boundary: Signed-but-out-of-bound calibration is rejected.
- Rollback-safe update: Interrupted update shall recover safely and unauthorized rollback shall be blocked. Acceptance / boundary: No bricked unsafe state; gate remains disabled until authenticated runnable image is present.

Algorithm / dynamic behavior

Secure startup/update: authenticate -> version -> calibration signature/range/variant -> handover; failure holds independent torque/gate enable disabled.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B15-01	Secure boot	Unsigned or modified production software shall be blocked and torque/gate enable shall remain disabled.	Negative security test blocks unsigned image.	SW	Source-derived	MR-TRC-016
TRC220-SYS3-B15-02	Safety calibration protection	Unauthorized changes to	Signed-but-out-of-bound calibration	SW	Source-derived	MR-TRC-016

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		safety/torque/gate calibration shall be rejected using signature/authentication plus immutable hard bounds.	is rejected.			
TRC220-SYS3-B15-03	Rollback-safe update	Interrupted update shall recover safely and unauthorized rollback shall be blocked.	No bricked unsafe state; gate remains disabled until authenticated runnable image is present.	SW	Engineering-derived	MR-TRC-016
TRC220-SYS3-B15-04	Variant coding	Software/ calibration shall match motor, inverter power module, voltage class and vehicle variant before torque enable.	Wrong-variant validly signed image/cal is rejected.	HW/SW	Engineering-derived	MR-TRC-016
TRC220-SYS3-B15-05	Key/HSM protection	Security keys shall use approved HSM/secure storage and production provisioning with genealogy.	Unauthorized access blocked; key provisioning traceable.	HW/SW	Engineering-derived	MR-TRC-016

Safety, diagnostics and failure containment

- Secure boot: Negative security test blocks unsigned image.
- Safety calibration protection: Signed-but-out-of-bound calibration is rejected.
- Rollback-safe update: No bricked unsafe state; gate remains disabled until authenticated runnable image is present.
- Variant coding: Wrong-variant validly signed image/cal is rejected.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-GEN-02 - ISO/SAE 21434 road vehicles - cybersecurity engineering (ISO): source link

21. B16 - EMC, Grounding, PCB & Isolation Architecture

SiC switching-current loops, gate-driver return, sensing grounds, shielding, CM control, SI/PI and vehicle EMC behavior.

Architecture and boundaries

- Power-loop / gate-loop layout: Commutation loop, gate loop and Kelvin source/emitter paths shall be minimized and separated from precision sensing/control returns. Acceptance / boundary: DPT confirms bounded overshoot/ringing and no false turn-on.

- Return path / grounding: Current/voltage/resolver/digital interfaces shall have controlled return paths and shall not cross noisy reference discontinuities without defined stitching/transition. Acceptance / boundary: SI/PI/EMC review closes; no signal integrity failure in switching tests.
- Gate-slew EMC envelope: Released gate-drive profiles shall satisfy conducted/radiated emissions, immunity, motor dv/dt and device-stress constraints simultaneously. Acceptance / boundary: No production profile outside validated multidomain envelope.

Algorithm / dynamic behavior

Design governance: DPT + SI/PI + EMC results constrain gate resistance/profile, layout and filtering; production calibration cannot exceed validated envelope.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B16-01	Power-loop / gate-loop layout	Commutation loop, gate loop and Kelvin source/emitter paths shall be minimized and separated from precision sensing/control returns.	DPT confirms bounded overshoot/ringing and no false turn-on.	HW	Engineering-derived	MR-TRC-008
TRC220-SYS3-B16-02	Return path / grounding	Current/voltage/ resolver/digital interfaces shall have controlled return paths and shall not cross noisy reference discontinuities without defined stitching/transition.	SI/PI/EMC review closes; no signal integrity failure in switching tests.	HW	Engineering-derived	MR-TRC-008
TRC220-SYS3-B16-03	Gate-slew EMC envelope	Released gate-drive profiles shall satisfy conducted/radiated emissions, immunity, motor dv/dt and device-stress constraints simultaneously.	No production profile outside validated multidomain envelope.	HW/SW	Source-derived	MR-TRC-020
TRC220-SYS3-B16-04	Common-mode immunity	Resolver/current/ driver/ communication circuits shall withstand inverter common-mode transients without unsafe torque or gate event.	CM injection/radiated immunity produces no unsafe torque/PWM.	HW/SW	Engineering-derived	MR-TRC-008
TRC220-SYS3-B16-05	EMC test modes	Every production PWM/modulation/ gate profile capable of field use shall be represented in EMC DVP.	No untested production switching mode.	HW/SW	Engineering-derived	MR-TRC-008

Safety, diagnostics and failure containment

- Power-loop / gate-loop layout: DPT confirms bounded overshoot/ringing and no false turn-on.
- Return path / grounding: SI/PI/EMC review closes; no signal integrity failure in switching tests.
- Gate-slew EMC envelope: No production profile outside validated multidomain envelope.
- Common-mode immunity: CM injection/radiated immunity produces no unsafe torque/PWM.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-08 - High-voltage traction inverter system solution (Infineon): source link

22. B17 - Mechanical, Cooling, Environmental & Reliability

Liquid-cooled mechanical stack, sealing, vibration/shock/ingress/altitude and 15-year/300,000 km lifetime margin.

Architecture and boundaries

- Environmental mission: Inverter shall meet installation temperature, vibration, shock, ingress and altitude mission profile. Acceptance / boundary: Environmental DVP completes without unsafe degradation/intermittent gate/sensor fault.
- Design life: Sensing, gate-drive, power module, DC-link and thermal margins shall support 15-year/300,000 km aging assumptions. Acceptance / boundary: Mission-profile lifetime analysis and endurance evidence close.
- Cooling stack: Cooler/TIM/module clamping/manifold shall maintain thermal resistance and leak integrity across life and mechanical exposure. Acceptance / boundary: Thermal impedance and leak/pressure tests remain within release limits after aging.

Algorithm / dynamic behavior

Reliability governance: mission profile -> thermal cycling/loss -> component damage accumulation -> derating/health flags -> DVP/release evidence.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B17-01	Environmental mission	Inverter shall meet installation temperature, vibration, shock, ingress and altitude mission profile.	Environmental DVP completes without unsafe degradation/intermittent gate/sensor fault.	HW	Source-derived	MR-TRC-026
TRC220-SYS3-B17-02	Design life	Sensing, gate-drive, power module, DC-link and thermal margins shall support 15-year/300,000 km aging assumptions.	Mission-profile lifetime analysis and endurance evidence close.	HW	Source-derived	MR-TRC-027
TRC220-SYS3-B17-03	Cooling stack	Cooler/TIM/module clamping/manifold shall maintain thermal resistance and leak integrity across life and mechanical	Thermal impedance and leak/pressure tests remain within release limits after aging.	HW	Engineering-derived	MR-TRC-019;MR-TRC-027

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
		exposure.				
TRC220-SYS3-B17-04	150% BOM	Program shall maintain 150% BOM evidence for safety MCU, SiC module and gate-drive technology families.	Alternate branch includes HW/SW/safety/EMC/thermal/DVP impact, not drop-in assumption.	HW/SW	Source-derived	MR-TRC-028
TRC220-SYS3-B17-05	Supplier errata/PCN	Supplier errata and process/design changes affecting special characteristics shall trace to workaround/impact verification.	No unresolved safety-relevant errata at release.	HW/SW	Engineering-derived	MR-TRC-028

Safety, diagnostics and failure containment

- Cooling stack: Thermal impedance and leak/pressure tests remain within release limits after aging.
- Supplier errata/PCN: No unresolved safety-relevant errata at release.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-TRC-01 - Infineon EV traction inverter application and 800 V system solution (Infineon): source link

23. B18 - Manufacturing, EOL Calibration & Traceability

EOL current/resolver/Vdc calibration, gate diagnostics, isolation/continuity and serialized HW/SW/cal genealogy.

Architecture and boundaries

- EOL current calibration: Production shall calibrate phase-current gain/offset with released fixture/master reference and store controlled coefficients. Acceptance / boundary: Offsets/gains within released limits; fixture identity and drift state recorded.
- EOL resolver alignment: Resolver electrical offset/alignment shall be measured/stored and verified at EOL. Acceptance / boundary: Angle offset within released limit; missing/invalid value blocks torque enable.
- EOL gate diagnostics: Gate-driver BIST/SPI/status, UVLO/fault reporting and safe-disable functionality shall be tested per channel. Acceptance / boundary: All required driver channels and independent inhibit pass.

Algorithm / dynamic behavior

EOL flow: identity -> flash/security -> sensor/resolver calibration -> gate diagnostics -> isolation/continuity -> record fixture/master -> genealogy release.

Detailed SYS.3 requirements

Req ID	Name	Requirement Statement	Acceptance / Boundary	Alloc.	Origin	Source MR
TRC220-SYS3-B18-01	EOL current calibration	Production shall calibrate phase-current gain/offset with released fixture/master reference and store controlled coefficients.	Offsets/gains within released limits; fixture identity and drift state recorded.	HW/SW	Source-derived	MR-TRC-030
TRC220-SYS3-B18-02	EOL resolver alignment	Resolver electrical offset/alignment shall be measured/stored and verified at EOL.	Angle offset within released limit; missing/invalid value blocks torque enable.	HW/SW	Source-derived	MR-TRC-030
TRC220-SYS3-B18-03	EOL gate diagnostics	Gate-driver BIST/SPI/status, UVLO/fault reporting and safe-disable functionality shall be tested per channel.	All required driver channels and independent inhibit pass.	HW/SW	Source-derived	MR-TRC-030
TRC220-SYS3-B18-04	EOL HV/isolation checks	Vdc measurement, HV isolation/continuity and production safe-state behavior shall be verified before release.	All released electrical/insulation thresholds pass.	HW/SW	Source-derived	MR-TRC-030
TRC220-SYS3-B18-05	Serialized traceability	Serial record shall link ECU/BOM/power module/driver/MCU/SW/calibration/security provisioning/EOL results and fixture IDs.	Genealogy complete before shipment; deviations closed.	SYS	Source-derived	MR-TRC-029;MR-TRC-030

Safety, diagnostics and failure containment

- EOL resolver alignment: Angle offset within released limit; missing/invalid value blocks torque enable.
- EOL gate diagnostics: All required driver channels and independent inhibit pass.
- EOL HV/isolation checks: All released electrical/insulation thresholds pass.

Verification / DVP focus

- nominal operating point
- minimum/maximum voltage corner
- hot/cold thermal corner
- sensor/communication fault injection
- startup/shutdown transition
- EMC/common-mode disturbance as applicable

Primary reference documents

- REF-GEN-04 - IATF 16949 automotive quality management (IATF): source link

24. Verification, DVP and Release Evidence Strategy

- MIL/SIL for control algorithms and limit managers before target integration.

- Target/HIL fault injection for sensor/network/CPU/safety-path timing.
- Bench double-pulse / power-stage / insulation / thermal tests as applicable.
- Power-analyzer/dyno/grid-simulator testing across released operating envelope.
- EMC, environmental, vibration, thermal shock, ingress and durability per vehicle DVP.
- EOL fixture correlation and serialized production evidence before SOP release.

25. Reference Document Register

External references provide design comparators, datasheets/application notes and standards context. They do not supersede the controlled product baseline. Freeze the exact revision/PDF in the project repository before release.

Ref ID	Block	Category	Document / Title	Source	Link	Use
REF-GEN-01	General	Standard	ISO 26262 road vehicles - functional safety	ISO	https://www.iso.org/standard/68383.html	Functional safety lifecycle and safety mechanisms.
REF-GEN-02	General	Standard	ISO/SAE 21434 road vehicles - cybersecurity engineering	ISO	https://www.iso.org/standard/70918.html	Cybersecurity lifecycle/TARA.
REF-GEN-03	General	Standard	ISO 14229 unified diagnostic services	ISO	https://www.iso.org/standard/72439.html	UDS diagnostics.
REF-GEN-04	General	Other	IATF 16949 automotive quality management	IATF	https://www.iatfglobaloversight.org/iatf-16949/	Production quality and traceability context.
REF-TRC-01	B01/B02/B07/B09/B10/B12/B17	Application / Reference	Infineon EV traction inverter application and 800 V system solution	Infineon	https://www.infineon.com/application/automotive-traction-inverter	System topology, MCU, driver and 800 V traction comparator.
REF-TRC-02	B01/B02/B04/B05/B09	Reference Design	TIDM-02014 800 V 300 kW SiC traction inverter reference design	Texas Instruments / Wolfspeed	https://www.ti.com/tool/TIDM-02014	Power stage, sensing, control and gate-drive comparator above 220 kW peak class.
REF-TRC-03	B03/B08/B11	Application Note	AURIX TC3xx main inverter functional safety reference	Infineon	https://documentation.infineon.com/aurixtc3xx/docs/nnc1745576047829	MCU, resolver, PWM and inverter safety architecture reference.
REF-TRC-04	B03	Product / Application	Automotive isolated gate drivers for traction inverters	Infineon	https://www.infineon.com/products/power/gate-driver-ics/automotive	Gate-driver isolation/protection reference.
REF-TRC-05	B06	White Paper / Design	Design priorities in EV traction inverters - resolver/front-end considerations	Texas Instruments	https://www.ti.com/lit/wp/sprad58a/sprad58a.pdf	Resolver and traction control design comparator.
REF-TRC-06	B08/B11	Reference Design	EV Traction Inverter Control Reference Design Gen 3	NXP	https://www.nxp.com/design/design-center/development-boards-and-designs/EV-INVERTERGEN3	800 V SiC control reference with ASIL D-oriented system collateral.
REF-TRC-07	B10	Application Note	AM263x for traction inverters / real-time motor control	Texas Instruments	https://www.ti.com/lit/an/sprad20/sprad20.pdf	FOC/control implementation reference.
REF-TRC-08	B16	Application Presentation	High-voltage traction inverter	Infineon	https://www.infineon.com	System/PCB/gate/thermal reference

Ref ID	Block	Category	Document / Title	Source	Link	Use
			system solution		/dgdl/Infineon-Infineon_High_voltage_traction_inverter_presentation-ApplicationPresentation-v05_00-EN.pdf?fileId=8ac78c8c8afe5bd0018b291672a47e70	for 800 V traction inverter.

Clickable reference links

- REF-GEN-01 - ISO 26262 road vehicles - functional safety: https://www.iso.org/standard/68383.html
- REF-GEN-02 - ISO/SAE 21434 road vehicles - cybersecurity engineering: https://www.iso.org/standard/70918.html
- REF-GEN-03 - ISO 14229 unified diagnostic services: https://www.iso.org/standard/72439.html
- REF-GEN-04 - IATF 16949 automotive quality management: https://www.iatfglobaloversight.org/iatf-16949/
- REF-TRC-01 - Infineon EV traction inverter application and 800 V system solution: https://www.infineon.com/application/automotive-traction-inverter
- REF-TRC-02 - TIDM-02014 800 V 300 kW SiC traction inverter reference design: https://www.ti.com/tool/TIDM-02014
- REF-TRC-03 - AURIX TC3xx main inverter functional safety reference: https://documentation.infineon.com/aurixtc3xx/docs/nnc1745576047829
- REF-TRC-04 - Automotive isolated gate drivers for traction inverters: https://www.infineon.com/products/power/gate-driver-ics/automotive
- REF-TRC-05 - Design priorities in EV traction inverters - resolver/front-end considerations: https://www.ti.com/lit/wp/sprad58a/sprad58a.pdf
- REF-TRC-06 - EV Traction Inverter Control Reference Design Gen 3: https://www.nxp.com/design/design-center/development-boards-and-designs/EV-INVERTERGEN3
- REF-TRC-07 - AM263x for traction inverters / real-time motor control: https://www.ti.com/lit/an/sprad20/sprad20.pdf
- REF-TRC-08 - High-voltage traction inverter system solution: https://www.infineon.com/dgdl/Infineon-Infineon_High_voltage_traction_inverter_presentation-ApplicationPresentation-v05_00-EN.pdf?fileId=8ac78c8c8afe5bd0018b291672a47e70

26. Appendix - Release Checklist

- Requirements review and architecture rationale approved.
- HWE.1/SWE.1 handoff and HSI timing/validity/owner defined.
- WCCA, DFMEA/DRBFM, safety/cyber analyses updated where applicable.
- All planned DVP cases have approved procedures and controlled equipment/configuration.
- All test failures have closed problem/deviation records and residual-risk disposition.
- RTM provides bidirectional link from source requirement through SYS.3/HWE.1/SWE.1 to verification and validation evidence.